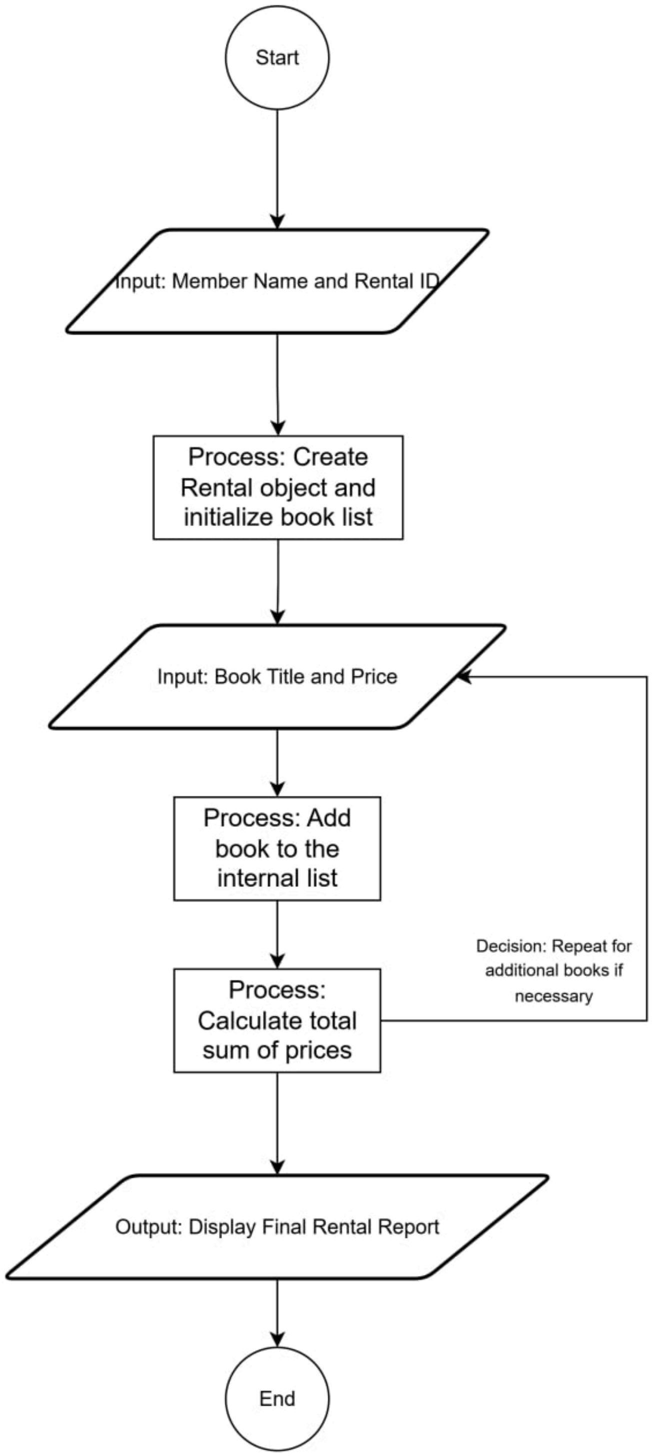
# Assignment: Community BookHub Rental Tracker

## Task 1: Algorithm Design (Logic Planning)

1. **Start:** Begin the program execution.
2. **Input Member Details:** Prompt the user to enter the member's name and a unique Rental ID.
3. **Initialize Object:** Create a new Rental object instance using the provided name and ID, initializing an empty list for books.
4. **Input Book Data:** Prompt the user to enter the title and rental price of a book.
5. **Add Book:** Append the book's details (title and price) to the object's book list.
6. **Check for More Books:** Determine if there are additional books to be added. If yes, repeat steps 4 and 5.
7. **Calculate Total:** Iterate through the book list and sum the prices to calculate the total rental cost.
8. **Display Report:** Generate and print a full summary report including the member name, ID, all rented books, and the total cost.
9. **End:** Terminate the program.

# Task 2: Logic Flowchart (Visual Representation)



# Task 3: Implementation (Python OOP Code )

```python
# Assignment: Community BookHub Rental Tracker

class Rental:
    def __init__(self, member_name, rental_id):
        # Set up member details and an empty list for books
        self.member_name = member_name
        self.rental_id = rental_id
        self.books = []

    def add_book(self, title, price):
        # Add book title and price to the list
        self.books.append({"title": title, "price": price})

    def calculate_total(self):
        # Add up all the prices in the book list
        total = sum(book['price'] for book in self.books)
        return total

    def display_report(self):
        # Print the final report with all details
        print("\n--- COMMUNITY BOOKHUB RENTAL REPORT ---")
        print(f"Member Name : {self.member_name}")
        print(f"Rental ID   : {self.rental_id}")
        print("-" * 30)

        for book in self.books:
            print(f"- {book['title']}: ${book['price']}")

        print("-" * 30)
        # Call the total method and show the result
        print(f"TOTAL COST: ${self.calculate_total()}")

# --- Program Start ---

def main():
    # Get user input for member info
    name = input("Enter Member Name: ")
    rid = input("Enter Rental ID: ")

    # Create the object
    my_rental = Rental(name, rid)

    # Loop to let the user add multiple books
    while True:
        book_title = input("\nEnter Book Title (or 'done' to stop): ")
        if book_title.lower() == 'done':
            break

        book_price = float(input(f"Enter price for {book_title}: "))
        # Add the book data to the object
        my_rental.add_book(book_title, book_price)

    # Show the final report
    my_rental.display_report()

if __name__ == "__main__":
    main()
```